CHALLENGE #5

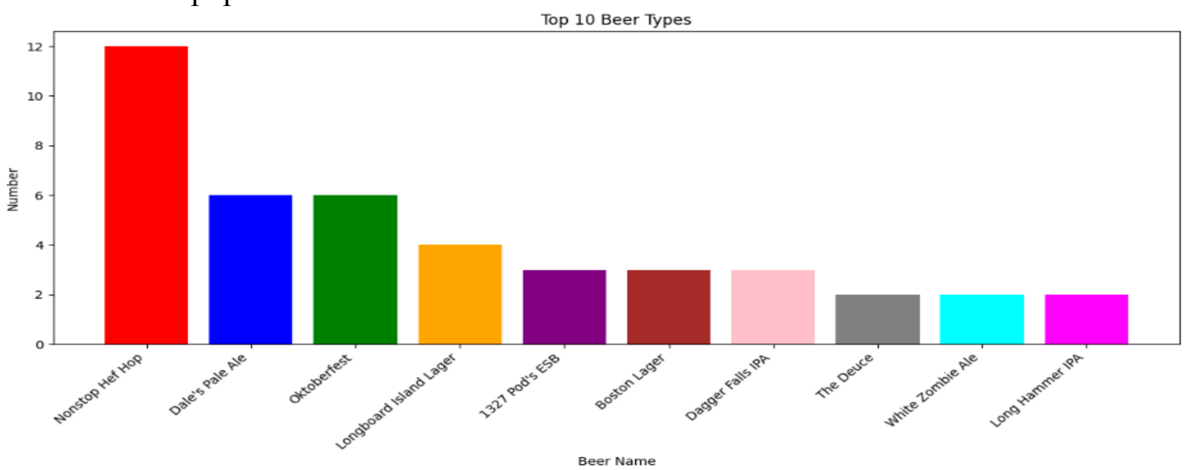
1. **Download Dataset**
We use the Python programming language with the Google Colab platform, so the dataset from Kaggle can be directly accessed because it is open source.
This dataset consists of two files, namely beer data and breweries data. Each dataset has different features along with the features and feature descriptions:
Beer Dataset: Name (Name of the beer), Style (Beer style), Abv (The alcoholic content by volume with 0 being no alcohol and 1 being pure alcohol), Brewery_id (Unique identifier for brewery that produced this beer), Ounces (Size of beer in ounces), Ibu (International bittering units, which describe how bitter a drink is)
Breweries Dataset: Brewery_id (Unique identifier for the brewery that produced this beer), Name (Name of the brewery), City (City that the brewery is in), State (State that the brewery is located).

2. **Some Questions from the Dataset**

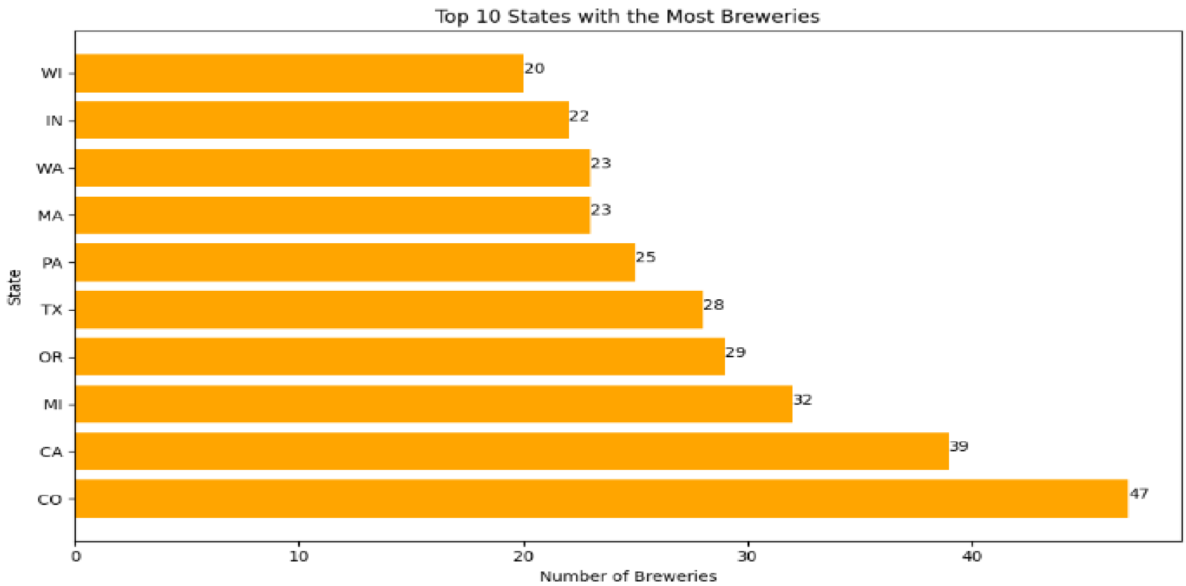
- What is the most popular beer?
- What countries have the most breweries?
- What beer styles are popular and what is the average alcohol content?
- Are some beers very bitter & some have high alcohol content?
- How is the Distribution of Alcohol Content in Beer

3. **Exploratory charts**

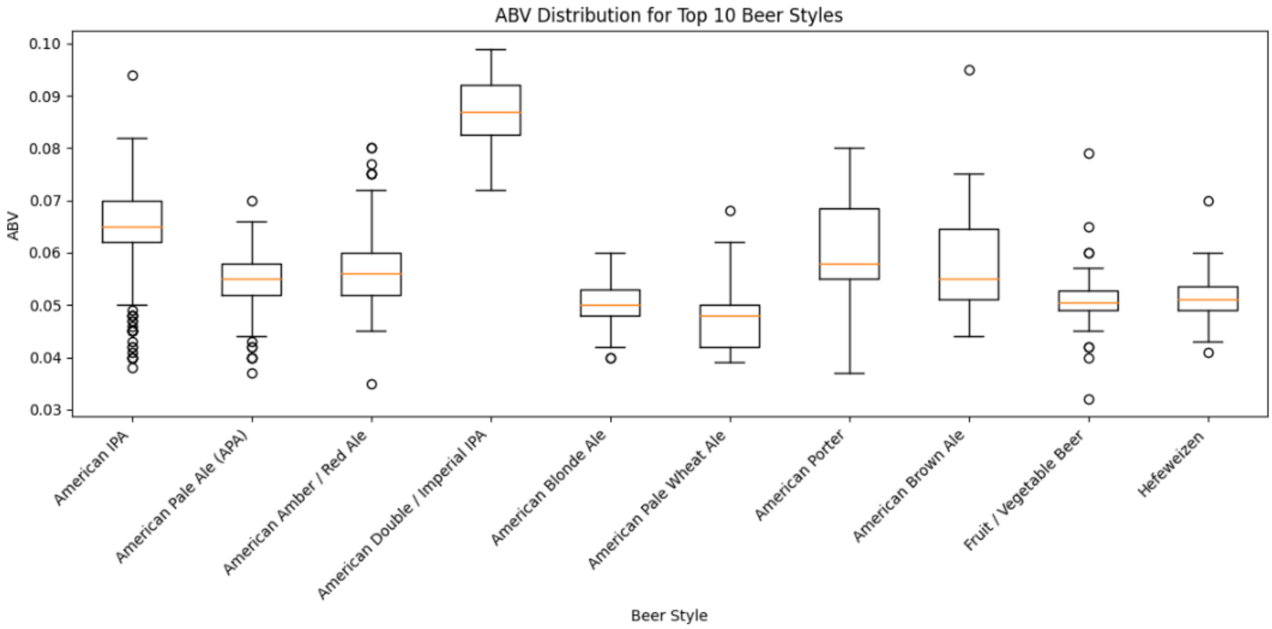
- What is the most popular beer?



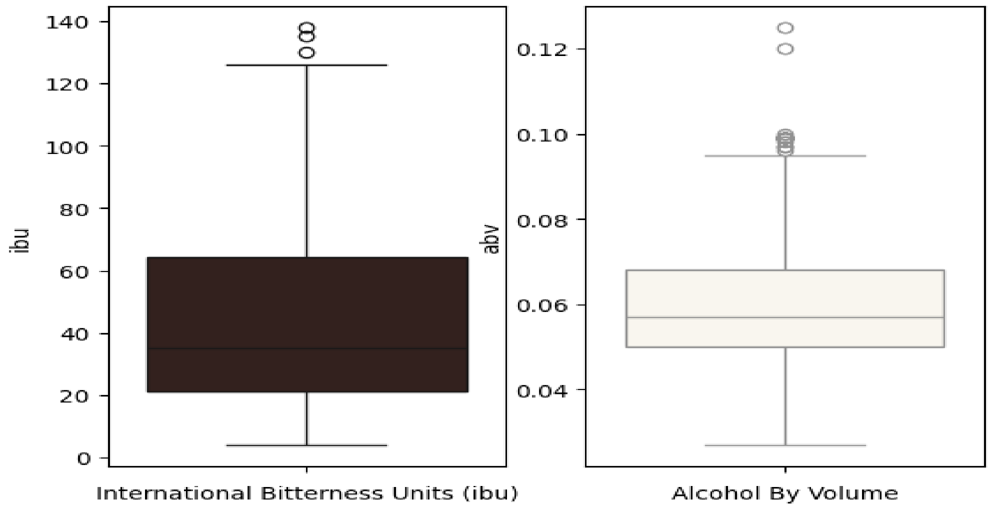
- What countries have the most breweries?



- What styles of beer are popular and what is their average alcohol content?



- Are some Beers very bitter & some have a high alcohol content?



4. Findings dan Insigths

- The most popular beer type is Nonstop Hef Hop.
- The state with the most breweries is Colorado, which has 47 breweries.
- The most popular beer style is American IPA, with an average alcohol content of 0.065 (6.5%).
- Most beers have a bitterness level (IBU) of around 40 and an alcohol level (ABV) of around 6%. The range of IBU values was more varied than the ABV, indicating more significant variation in bitterness than alcohol content in the analysed beers.

5. Insights

In the first analysis, we found that nonstop hef hop is the most popular beer, so this product will be the strongest competitor when we make beer products.

The second analysis then reveals some places with the most breweries, which we can visit to find beer suppliers or learn about beer types.

Furthermore, the third and fourth analyses provide more specific information, namely the most popular beer styles and their alcohol content, which can be our basis when making beer that can compete in the beer market.